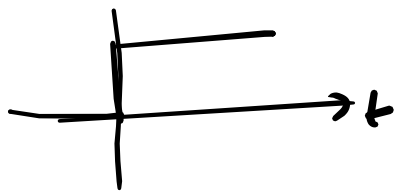
12/11/2025

C.T. F.S.

$$\Rightarrow \hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$\rightarrow a_k = \langle x(t), \phi_k(t) \rangle_T = \frac{1}{T}\int_T x(t)\,\phi_k^*(t)\,dt$$

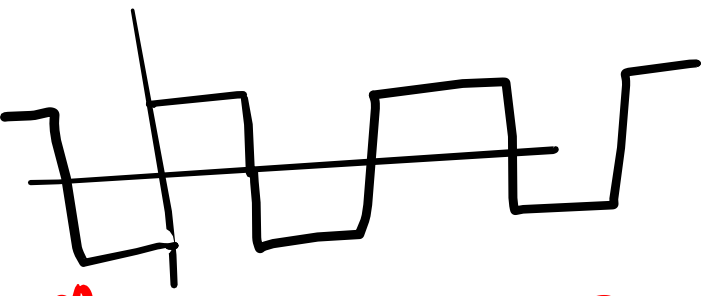
$$= \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\,dt$$



$$\hat{x}_1(t) = \sum_{k=-1}^{1} a_k \phi_k(t)$$

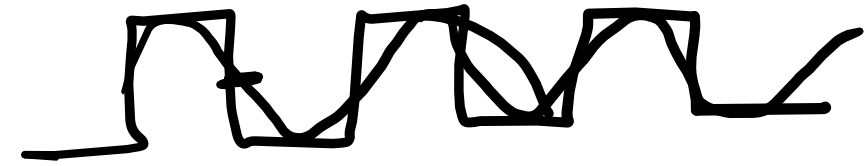
$$\hat{x}_2(t) = \sum_{k=-10}^{10} a_k \phi_k(t)$$

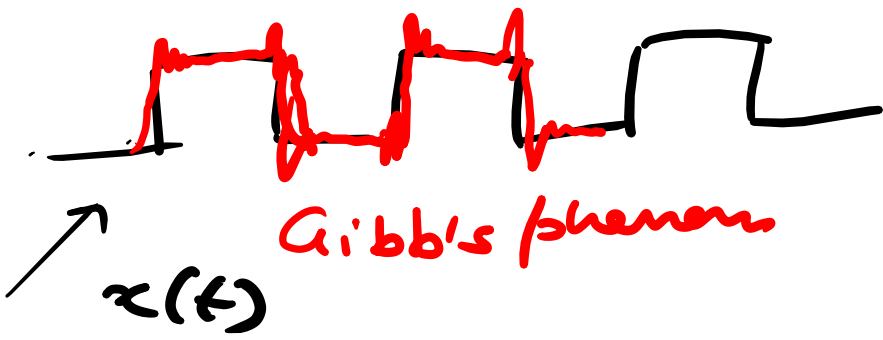
$$\hat{x}_n(t) \longrightarrow x(t)$$



$$\int_{-\infty}^{\infty} |\hat{x}(t) - x(t)|^2 dt \rightarrow 0$$

$\hat{x}(t) = x(t)$ a.e.





Laplace Transform

$$X(s) \triangleq \int_{-\infty}^{\infty} x(t) e^{-st}\, dt$$

Specify ROC.

Ex#1 $x(t) = e^{-at} u(t)$

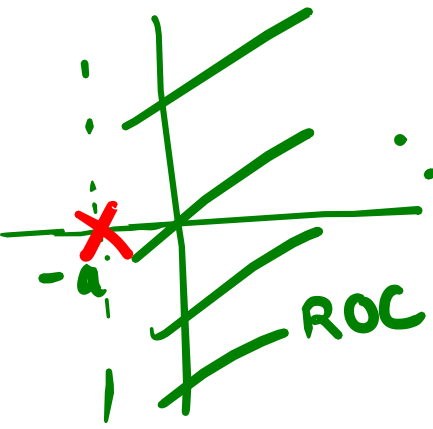
$\downarrow$

$$X(s) = \frac{1}{s+a}; \quad \underbrace{\mathrm{Re}\{s\} > -a}_{\text{ROC}}$$

if $a > 0$

$s = -a$ is called the pole of $X(s)$

$\therefore X(s) \to \infty$ at $s = -a$

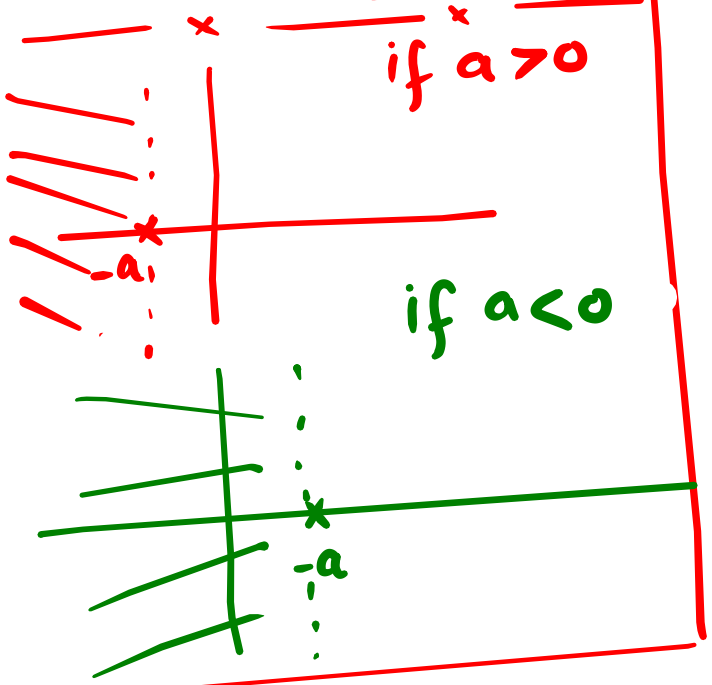


Ex#2 $x(t) = -e^{-at} u(-t)$; $a > 0$

$\sigma < -a$

This implies

$\mathrm{Re}\{s\} < -a$



$$X(s) = \int_{-\infty}^{0} -e^{-at}u(-t)e^{-st}dt$$

$$= -\int_{-\infty}^{0} e^{-(s+a)t}dt = \frac{-e^{-(s+a)t}}{-(s+a)}\Bigg|_{-\infty}^{0}$$

$$= \frac{e^{-(s+a)t}}{t(s+a)}\Bigg|_{t=-\infty}^{0} = \frac{1}{s+a} - \frac{e^{-(\sigma+j\omega+a)t}}{s+a}\Bigg|_{t=-\infty}$$

$$= \frac{1}{s+a} - \frac{e^{t(\sigma+a)\infty} \cdot e^{j\omega\infty}}{s+a}$$

$$X(s) = \frac{1}{s+a}$$

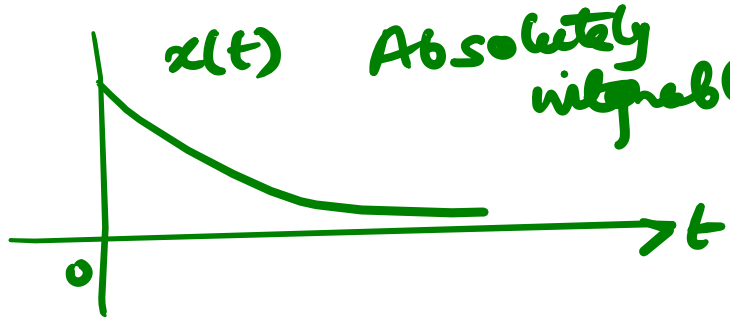
ROC: $\mathrm{Re}\{s\} < -a$

For $X(s)$ to be convergent (not going to infinity), $e^{(\sigma+a)\infty}$ should tend to zero. This will happen if $(\sigma+a) < 0$

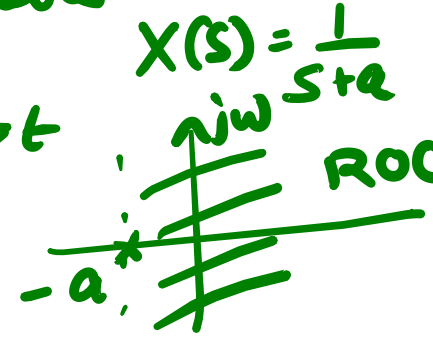
$\sigma < -a$

$e^{-at}u(t)$

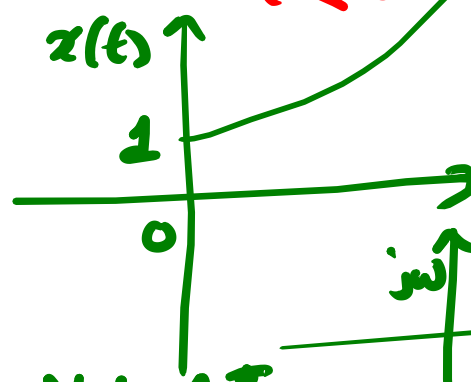
a > 0



$X(s) = \frac{1}{s+a}$



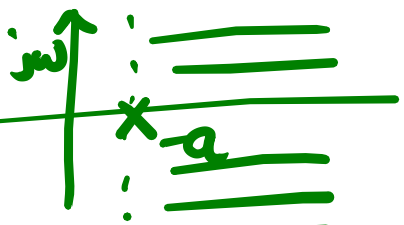
a < 0



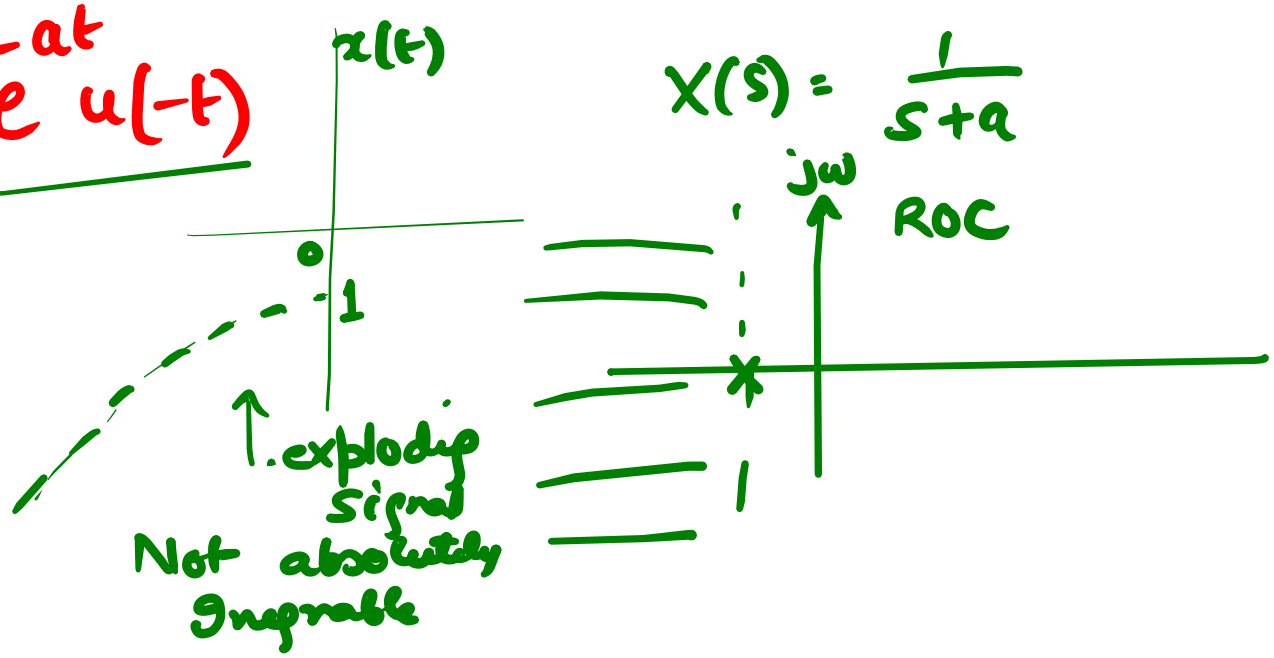
Not. A.I
Not. F.T.

$X(s) = \frac{1}{s+a}$

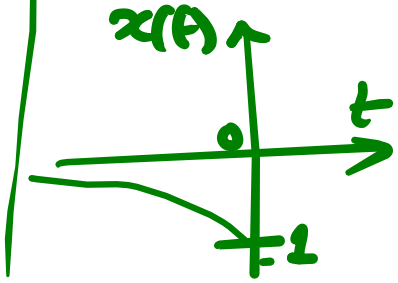
$Re\{s\} > -a$



$-e^{-at}u(-t)$



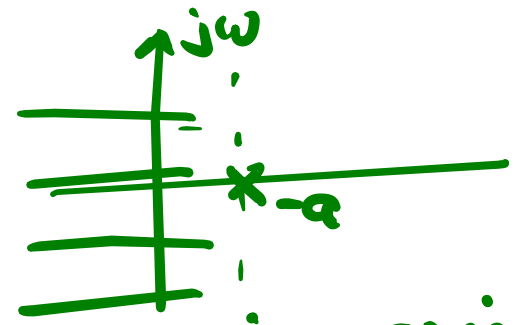
$X(s) = \frac{1}{s+a}$



A.I.
F.T. exists

$X(s) = \frac{1}{s+a}$

$Re\{s\} < -a$



ROC includes jw-axis

$$x(t) = e^{-b|t|} = e^{-bt}u(t) + e^{bt}u(-t)$$

① ②

Ex

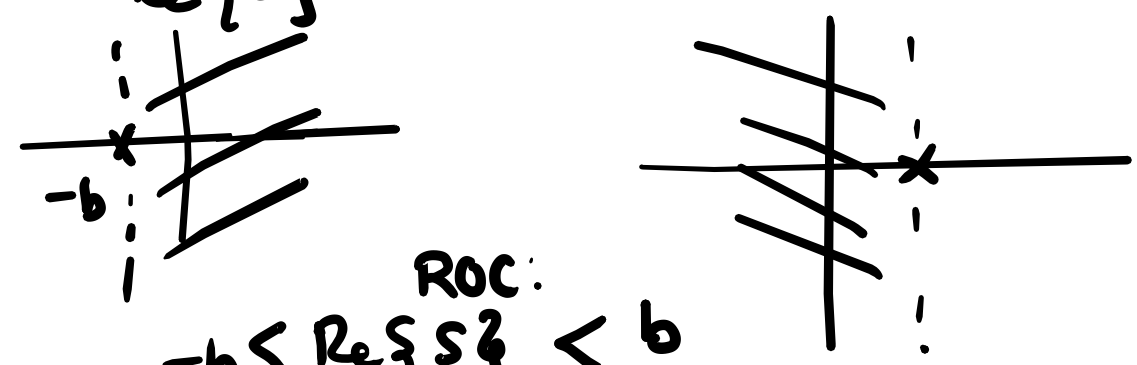
Two cases. (i) $b > 0$

(ii) $b < 0$

$\frac{1}{s+b}$ $\qquad$ $\frac{-1}{s-b}$

$b>0$

$\mathrm{Re}\{s\} > -b$ $\qquad$ $\mathrm{Re}\{s\} < b$



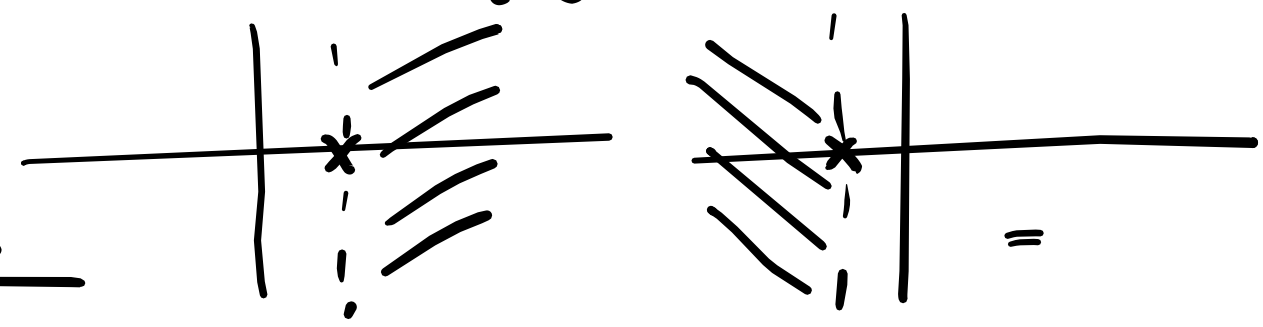
ROC:

$-b < \mathrm{Re}\{s\} < b$

$b<0$

Laplace does not exist

No ROC



Properties

① Linearity

$x_1(t) \rightleftharpoons X_1(s)$ ROC : R_1

$x_2(t) \rightleftharpoons X_2(s)$ ROC : R_2

$x_1(t) + x_2(t) \rightleftharpoons X_1(s) + X_2(s)$ ROC : $R_1 \cap R_2$

or may be longer.

② Time Shifting

$x(t) \rightleftharpoons X(s)$ ROC : R_1

$y(t) = x(t - t_0) \rightleftharpoons e^{-st_0} X(s)$ ROC : R_1

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$= e^{-st_0} X(s)$$

$t - t_0 = \tau \;\Big|\; t = \tau + t_0$

$dt = d\tau$

$$Y(s) = \int_{-\infty}^{\infty} x(t - t_0)\, e^{-st}\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau)\, e^{-st_0} \cdot e^{-s\tau}\, d\tau$$

$$= e^{-st_0} X(s)$$